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Class-02-28-Mar-Intro-GenAI-Tools-UV-Setup-Course-Access-&-Resources-Guide-Git-&-Github-Setup-Guide-by-Sunny

🔹 Install UV (if not installed)

```
pip install uv

if you are using MAC: pip3 install uv
```

🔹 Check UV installed.

```
uv --version
```

🔹 Check Installed Packages

```
optional  command: uv pip list
```

🔹 Check Available Python Versions

```
uv python list

if you wanted to install any specific version of the python using the uv then
use the below command:

uv python install <mention the cpython interpreter version>
```

🔹 Create Virtual Environment

Specific version:

```
uv venv <your-env-name> --python <your-python-version>
```

From requirements file:

```
uv pip install -r requirements.txt
```

Course Access & Resources Guide

1. **Log in to the platform:** https://krishnaikacademy.com/
2. **Access your course:** Navigate to your enrolled course and check all the prerequisites and live class recordings.
3. **Direct course portal:** https://learn.krishnaikacademy.com/web/courses/
4. **GitHub Repository (All Resources + Notion Notes):** https://github.com/sunnysavita10/Full-Stack-GenAI-Bootcamp-1.0
5. **Notice Board (Google Sheet):** https://docs.google.com/spreadsheets/d/1M7Dyr5EjDwu9EHIL5hvSJBmIQXifGH8QavIo_tbh8gk/edit?usp=sharing

Git & GitHub Setup Guide (Step-by-Step)

✅ 1. Install Required Tools

Make sure you have the following installed:

- Visual Studio Code
- Git
- GitHub account

Check Git installation:

```
git --version
```

📁 2. Create a New Project Folder

Method 1: Create locally and link to GitHub

Option A (Using VS Code)

- Open VS Code
- Click **File** → **Open Folder**
- Create a new folder (example: `my-project`)

Option B (Using Terminal / CMD)

```
mkdir my-project
cd my-project
code .
```

🔧 3. Initialize Git in Your Project

Inside the VS Code terminal:

```
git init
```

This creates a hidden `.git` folder and starts tracking your project.

📄 4. Create Project Files

Example:

- `main.py`
- `requirements.txt`

➕ 5. Add Files to Git

```
git add .
```

This tracks all files in the folder.

💾 6. Commit Your Code

```
git commit -m "Initial commit"
```

This saves a project snapshot.

🌐 7. Create a Repository on GitHub

- Go to GitHub
- Click **New repository.**
- Give it a name (usually the same as the folder's).
- Click **Create repository.**

🔗 8. Link Local Project to GitHub

Copy your repository URL (example):
`https://github.com/yourname/my-project.git`

Then run:

```
git remote add origin https://github.com/yourname/my-project.git
```

🚀 9. Push Code to GitHub

```
git branch -M main
git push -u origin main
```

🔁 **Alternative Method (Clone First)**

1. Create a new repository on GitHub
2. Copy the repository URL
3. Clone the repository

```
git clone https://github.com/yourname/my-project.git
```

This downloads the repo into your current directory.

1. Open the project in VS Code
2. Make changes (add files, notebooks, folders)
3. Add changes

```
git add .
```

1. Commit changes

```
git commit -m "your message"
```

1. Push to GitHub

```
git push
```

LLM / GenAI Development: Tools & Frameworks

LLM Foundations

- Huggingface
- PyTorch

Finetuning

- Unsloth
- LLaMA Factory
- Huggingface

Vector Databases & Search (RAG Backbone)

- FAISS

- Chroma
- Qdrant
- Supabase
- pgvector
- Milvus
- Pinecone
- AWS OpenSearch
- Azure AI Search

Databases & Storage

- Neo4j
- MongoDB
- PostgreSQL
- Redis
- SQL Alchemy

LLM Application Building Framework

- LangChain
- Llama Index
- Haystack
- Lang Smith

Guardrails, Validation & Safety

- Guardrails.ai
- Nvidia Nemo
- OpenAI Guardrail

Local LLMs & Runtime

- Ollama

Experiment Tracking

- MLflow

Data Modelling

- Pydantic

MCP

- FastMCP

Document AI & Parsing

- Llama Parser
- Docling
- Google Document AI
- Amazon Textract
- python-docx
- PyMuPDF
- pdfplumber
- Parsio
- Unstructured.io
- Azure Document Intelligence

Agentic AI Frameworks

- Lang Graph
- Deepagent
- AutoGen
- n8n

Cloud GenAI Platforms

- Google ADK
- Agent Core
- AWS Bedrock
- Azure Cloud Foundry
- GCP Vertex AI

Evaluation & Observability

- RAGAS
- TruLens
- Llama Index Eval
- DeepEval
- OpenAI Evals

Prompt Engineering & Prompt Management

- LangChain Prompt Hub

- PromptLayer
- OpenAI Cookbook / Prompt Examples
- PromptHub
- DSPy (Prompt-as-Code approach)
- Guidance (Microsoft)

Deployment, Infrastructure & CI/CD

- ECS + Fargate + EC2
- AWS SageMaker
- GitHub Actions
- Azure Virtual Machines (VMs)
- Azure Kubernetes Service (AKS)
- Azure Container Apps / ACI

Testing

- pytest

Pipeline Orchestration

- Apache Airflow